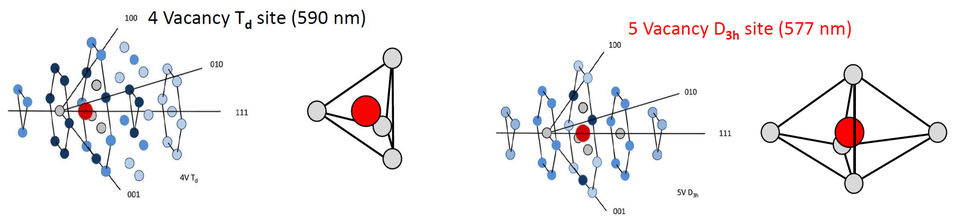
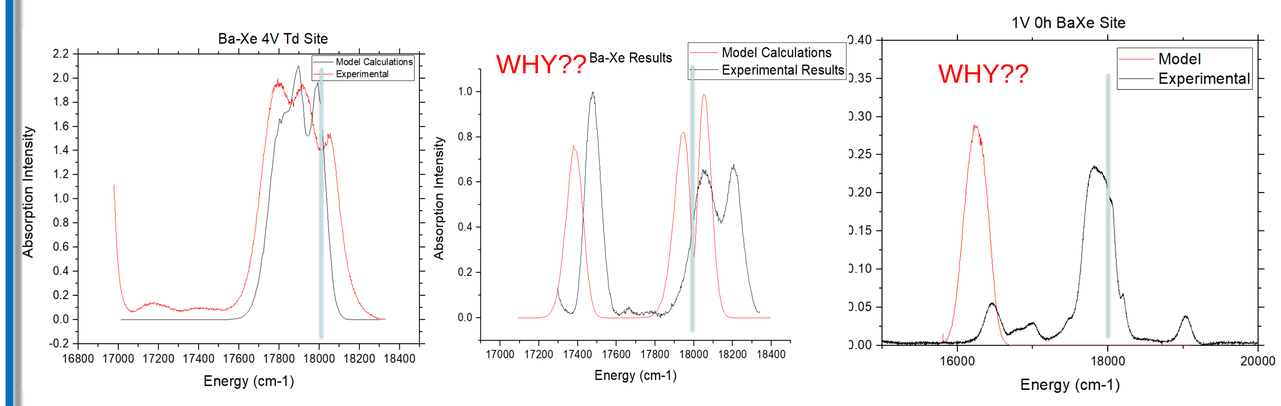
Modelling visible spectroscopy of the alkaline earth metals in rare gas matrix

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Abstract : The following work attempts to present a theoretical model using Diatomic in Molecule (DIM) to perform molecular dynamics on alkaline earth metals (Ca, Sr or Ba) inside the Rare Gas (RG) matrix (Ar, Kr or Xe) to investigate formation and properties of different trapping sites (1-Single Vacancy{V}-O_h, 4V-T_d and 5V-D_{3h}) considering a complete two-electron problem for orbitals ns, np and $(n-1)d$ for $n = 4,5,6$ in Ca, Sr and Ba. We intend to investigate the unusual D_{3h} trapping site for Ba by considering quadrupole polarizability.

Theoretical Model

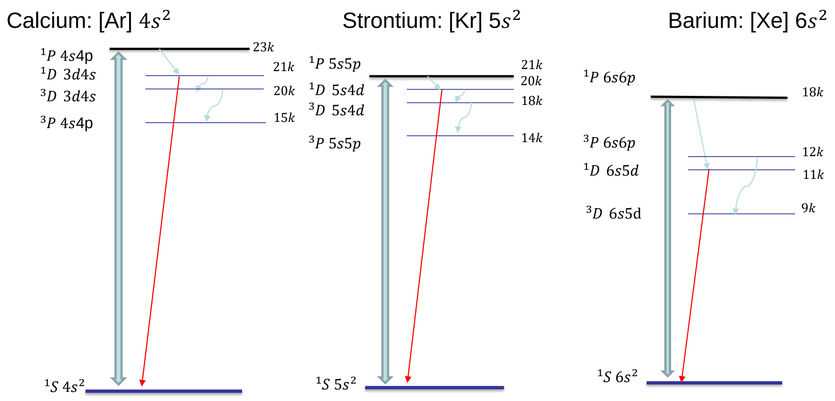




<u>Shift and broadening of peaks:</u>
- Non Zero temperature
- Static deformation
- Dynamic deformation

<u>Additional splitting:</u> [1]
- Non-degenerate 3-p shells and each state fits better in one of the directions: Jahn-Teller 2+1 splitting

<u>Abnormalities</u>:
- Too red shifted
- May improve by introduction of overlap and couplings

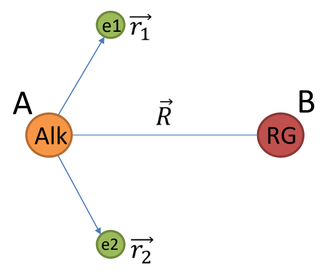


Spin Orbit Coupling allows some intrastate dipole forbidden transitions
Symmetry breaking in matrix allows other forbidden transitions

- **Steps to perform dynamics**
  - Generation of trapping site
    - Selection of a RG matrix (FCC or grain boundary) approximately 600 atoms
  - Introduce impurity in a desired cavity
  - Relaxation of system with reference to ground state of RG until the equilibrium geometry is achieved using damped molecular dynamics
  - Calculation of vibrational modes by diagonalization of Hessian matrix obtained from the equilibrium positions
  - We obtain the spectrum from these vibrational modes according to Boltzmann distribution
- **Parameterization**
  - with quantum chemistry program MOLCAS [2], we calculate dimer energies using Multi-Reference Configuration Interactions (MRCI)

Diatomic in Molecule (DIM) Model

- Based on valance bond theory (VBT) (Alkali and RG pure atomic orbitals) (exact requires $\sim 10^{6-9}$ configurations) [3]
- It uses significantly a reduced configuration space of VBT ($\sim 10^2$configs)
- Whole alkali-RG system $\hat{H}$ can be formulated from single alkali-RG diatomic adiabatic energies combined with rotations



$$|\Psi_{tot}\rangle = \hat{A}\,|\Psi_A\rangle \otimes |\Psi_B\rangle$$

$$|\Psi_A\rangle = \frac{1}{\sqrt{2}}\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$\hat{H} = \sum_B \left(\hat{H}_{AB} - \hat{H}_A\right) + \hat{H}_A + \sum_{B<B'} V_{BB'}$$

Diatomic | Alkaline Hamiltonian | Matrix Potential

$$\langle \Psi_m | \hat{H} | \Psi_n \rangle = H_{mn}$$

$$\hat{H}_A = \sum_{..} |n\rangle E_n^A \langle n| \xrightarrow{\text{Rotations}} \hat{H}_{A,pq} = \sum_{..} \langle p|n\rangle E_n^A \langle n|q\rangle$$

Atomic Frame → Lab Frame

$$\hat{H}_{AB,mn} = \sum_m |m\rangle E_m^{AB} \langle m| \longrightarrow \hat{H}_{AB,pq} = \langle p|\hat{H}_{AB}|q\rangle = \sum_m \langle p\,|\hat{R}^\dagger\,|m\rangle E_m^{AB} \langle m|\hat{R}\,|q\,\rangle$$

Molecular Frame

Conclusion

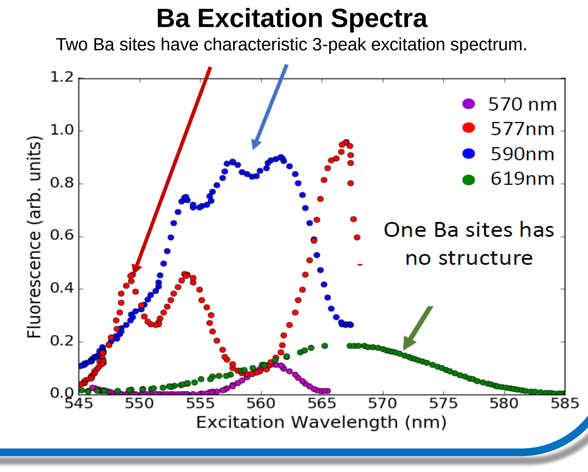
- From our elementary calculation, we understand that geometry plays a vital role in formation of trapping site
- We aim to extend the previously existing model to include 2 electrons and d-orbitals
- We expect that abnormal behaviour of 5V trapping site of Ba will be explained by considering low lying d-orbital
- We also anticipate to recover unwanted red shift of single vacancy site by considering overlap

Application

- n-EXO experiment, i.e. neutrinoless double beta decay [4]

$$^{136}Xe \rightarrow {}^{136}Ba^{2+} + 2e^-$$

- Help in identifying the possible sites to find Ba



Reference

[1] Davis, Gervais, and McCaffrey, (J. Chem. Phys. **148**, 124308 (2018)
[2] MOLCAS 7
[3] ***Ellison*** *J. Am. Chem. Soc.*1963,85,22,3540-3544
[4] **B. MONG et al**. PHYSICAL REVIEW A **91, 022505 (2015)**

Collaborations: